

Artificial Intelligence-Driven Malware Detection via Windows Event Log Engineering in Smart Grid Control Systems

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Motivation

The Security Gap in Smart Grid Hosts

Smart grids converge IP networks with Windows-based SCADA and RTU hosts, creating an IT/OT boundary actively exploited by adversaries — Stuxnet (2010), Ukrainian grid attacks (2015, 2016), Colonial Pipeline (2021).

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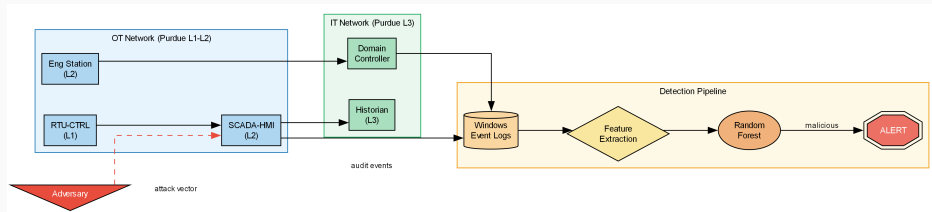
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Goal: apply machine learning to Windows Event Logs at Purdue Levels 1–3 to detect malicious host sessions in real time.

Smart Grid IT/OT Architecture



Background

Prior Work and Research Gap

Three research streams inform this work — each leaves a gap we fill:

Log anomaly detection (Drain, DeepLog, LogAnomaly) targets distributed IT logs — HDFS, syslog, OpenStack — not Windows Security Event IDs in OT.

ICS/SCADA IDS (Carcano, Goldenberg, SWaT dataset) monitors network traffic and physical process measurements, not host-level audit logs.

Windows malware detection (Achmad 2025, Önal 2025, Zhen 2025) focuses on IT endpoints, not OT hosts with ICS-specific event semantics.

Confirmed gap [1]

Host-level IDS is rare in the cyber-physical systems literature. No prior work targets Windows Event Logs at Purdue L1–L3 in smart grids. **This work is the first such framework.**

Threat Model

Threat Model: Adversary and Goals

Starting Position

Foothold on Level 3 IT host via phishing or supply-chain compromise. Valid or cracked domain credentials. Network access across a misconfigured IT/OT DMZ.

Attack Chain

Persist → Escalate privileges → Execute disruptive payload

(rogue PLC commands *or* denial of operator visibility)

Threat Model: MITRE ATT&CK ICS Patterns

Four attack categories modelled, each producing a distinct Windows Event Log signature:

1. **Lateral movement** — 3–8 consecutive Event 4625 logon failures from an unexpected source IP, followed by a 4624 success
2. **Persistence** — Event 7045 (service install) on a SCADA host, followed by 7036 (state change) and 4688 (process creation)
3. **Privilege escalation** — Event 4672 (special privileges) immediately preceding 4688 with an anomalous binary (`mimikatz.exe`, `psexec.exe`)
4. **Reconnaissance** — burst of 8–20 Event 4663 (object access) records within 30 seconds on HMI or RTU hosts

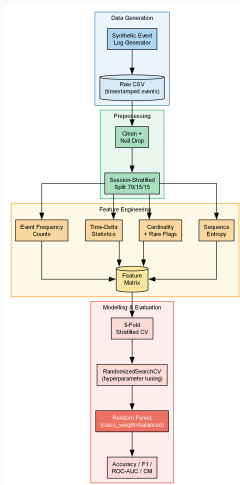
Data and Features

Synthetic Dataset: Justification and Statistics

No publicly available Windows Event Log dataset exists for smart grid SCADA or RTU environments [2], [3]. Synthetic generation is established practice: Goh et al. [4] built the SWaT testbed for the same reason; Chen et al. [5] used APT emulation when real logs were unavailable.

- **6,000 sessions:** 5,000 benign + 1,000 malicious (5:1 ratio)
- **89,288 total events** across 13 Windows Event IDs
- **7 smart grid hostnames:** SCADA-HMI-01, RTU-CTRL-03, eng_station, historian, ...
- **Evasive variants:** rate-limited lateral movement (60–300 s between failures) and slow reconnaissance (2–9 s burst intervals) prevent trivial single-feature separation

Data Pipeline: Generation to Evaluation



Session-Level Feature Engineering

Events grouped into sessions; each session produces a **24-dimensional vector** with no feature scaling required:

Group	Features	Dim.
Event frequency counts	count per event ID + total	14
Time-delta statistics	mean, std, min, max inter-event gap (seconds)	4
Host/user cardinality	unique sources, users, hostnames per session	3
Rare-event flags	binary: event in <5% of benign sessions	var
Sequence entropy	$H = -\sum_e p_e \log_2 p_e$	1

Design rationale

Mixed feature types handled natively by tree ensembles without normalisation. Session abstraction avoids deep sequence model latency while capturing temporal burst structure through time-delta statistics.

Results

Classification Results: Six-Method Comparison

Random Forest (class_weight=balanced, RandomizedSearchCV, 5-fold stratified CV)
versus five baselines:

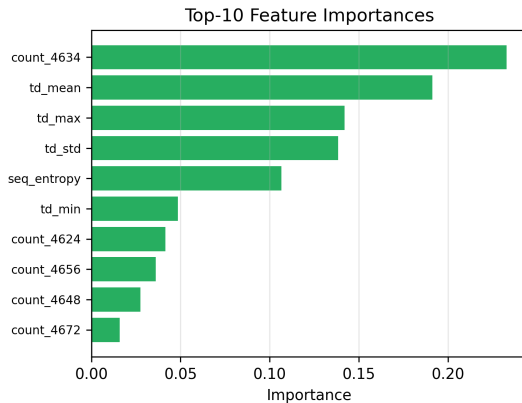
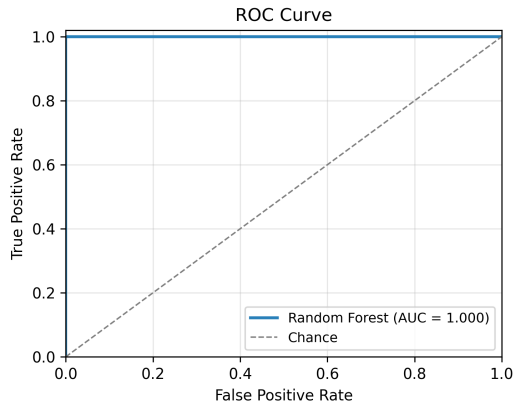
Method	Precision	Recall	F1 (macro)	ROC-AUC
Isolation Forest (unsupervised)	0.611	0.594	0.601	0.735
Logistic Regression	0.917	0.974	0.942	0.994
SVM (RBF kernel)	0.818	0.881	0.844	0.951
Decision Tree	0.982	0.987	0.984	0.988
XGBoost	0.997	0.999	0.998	1.000
Random Forest (proposed)	1.000	1.000	0.999	1.000

RF stability across 5 independent seeds: $F1 = 0.999 \pm 0.001$ · $AUC = 1.000 \pm 0.000$.
All metrics computed from the actual pipeline output.

No-label lower bound

Isolation Forest $F1 = 0.601$ shows
detection difficulty without any labelled
attack sessions.

ROC Curve and Feature Importances



Ablation Study: Feature Group Contribution

Removed Group	F1	$\Delta F1$
None (full model)	1.000	—
Group 1: Event counts	0.944	-0.056
Group 2: Time-delta stats	0.972	-0.028
Group 3: Cardinality	1.000	0.000
Group 4: Rare-event flags	1.000	0.000
Group 5: Entropy	1.000	0.000

Groups 3–5 show zero marginal delta because Groups 1 and 2 already fully separate the classes. Entropy alone achieves $F1 = 0.77$ in isolation — real signal, but redundant once counts and timing are present.

Practical takeaway

If timestamp resolution is unavailable, preserve event counts first ($\Delta F1 = -0.056$ on removal). On noisier real-world logs, Groups 3–5 are expected to recover importance.

Conclusion

Conclusion and Future Work

Contributions:

- First open-source framework for Windows Event Log malware detection in smart grid ICS (Purdue L1–L3)
- 24-feature session pipeline with evasive attack variants
- $F1 = 0.999 \pm 0.001$ (5 seeds); ROC-AUC = 1.000
- Event counts most discriminative ($\Delta F1 = -0.056$); timing statistics second (-0.028)
- Inference < 0.5 s per 900-session batch, single CPU core

Future Work

- Real ICS logs from red-team exercises
- Multi-class ATT&CK attribution
- Online learning for concept drift
- Adversarial robustness on real logs

- [1] R. Mitchell and I.-R. Chen, **“A survey of intrusion detection techniques for cyber-physical systems,”** *ACM Computing Surveys*, vol. 46, no. 4, pp. 1–29, 2014. DOI: 10.1145/2542049
- [2] M. Conti, D. Donadel, and F. Turrin, **“A survey on industrial control system testbeds and datasets for security research,”** *IEEE Communications Surveys & Tutorials*, vol. 23, no. 4, pp. 2248–2264, 2021. DOI: 10.1109/COMST.2021.3073482
- [3] M. N. Nafees, N. Saxena, A. A. Cardenas, S. Grijalva, and P. Burnap, **“Smart grid cyber-physical situational awareness of complex operational technology attacks: A review,”** *ACM Computing Surveys*, vol. 55, no. 10, pp. 1–36, 2023. DOI: 10.1145/3565570

- [4] J. Goh, S. Adepu, K. N. Junejo, and A. Mathur, **“A dataset to support research in the design of secure water treatment systems,”** in *Critical Infrastructure Protection X*, 2016, pp. 88–112. DOI: 10.1007/978-3-319-48737-3_5
- [5] C.-K. Chen et al., **“Building machine learning-based threat hunting system from scratch,”** *Digital Threats: Research and Practice*, vol. 3, no. 4, pp. 1–22, 2022. DOI: 10.1145/3559764

Thank you.

Questions welcome.